

DRPO Pipeline v2.3 Core: Evidence-First Vertical Slice

Anonymous review artifact

Scope. This two-page artifact validates the manuscript pipeline, not a new experiment. It uses the long-run-validated C-U1-E3-ADAM-RERUN compact repository evidence. C-U1 evaluates same-distribution held-out-context generalization. Task-performance collapse, support or variance-boundary events, and NaN/Inf numerical failures are reported separately.

Targeted Causal Transmission

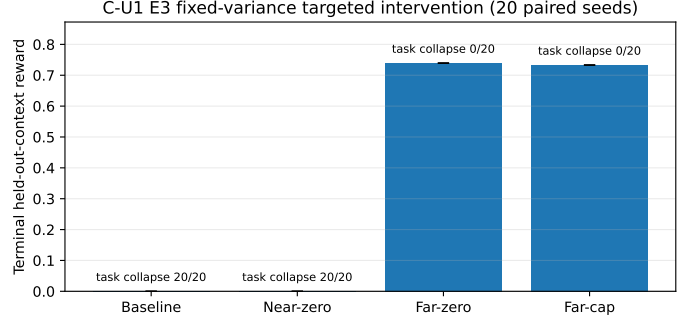


Table 1: Terminal outcomes. Task, support-boundary, and NaN/Inf events are separate.

Method	Fixed reward (95% CI)	Task	Support	NaN/Inf
Baseline	2.22e-06 [2.10e-06, 2.35e-06]	20/20	20/20	0/20
Near-zero	2.18e-06 [2.05e-06, 2.30e-06]	20/20	20/20	0/20
Far-zero	0.739 [0.739, 0.740]	0/20	0/20	0/20
Far-cap	0.733 [0.733, 0.734]	0/20	0/20	0/20
Global-scale	0.599 [0.599, 0.599]	0/20	0/20	0/20
Far-to-near	0.875 [0.875, 0.876]	0/20	–	0/20

Figure 1: Fixed-variance terminal reward with 95% CIs; labels show task-collapse counts.

We test the transmission path with four matched interventions over 20 paired held-out seeds. The uncontrolled Baseline and Near-zero variants finish at rewards 2.22e-06 and 2.18e-06 and undergo task-performance collapse in 20/20 and 20/20 seeds. Removing the far-field contribution instead yields 0.739 [0.739, 0.740] for Far-zero, while capping it yields 0.733 [0.733, 0.734] for Far-cap; neither intervention collapses in any seed. The registered fixed-variance budget controls also remain non-collapsed: Global-scale reaches 0.599, and transferring the far budget to the near component reaches 0.875. These controls are diagnostic and do not define a method ranking. Near-field removal is therefore not a rescue, whereas deleting or bounding the far-field path is. Within this controlled same-distribution held-out-context setting, the comparison identifies the far-field component as the dominant causal transmission path; it does not establish a universal method ranking.

The learnable-variance branch separates the type of failure. Baseline and Near-zero reach the registered support/variance-contraction boundary in 20/20 and 20/20 seeds, with mean onsets at steps 72.9 and 73.1. Far-zero and Far-cap record 0/20 and 0/20 support-boundary events. All four methods keep finite parameters and record 0/20 NaN/Inf failures. Thus this branch is evidence for support contraction rather than numerical collapse, and the intervention again isolates far-field negative influence as the removable path in C-U1.

Tail-Control Guarantee

Proposition 1 (Vanishing weighted far-field gradient). *Let $r \geq 0$ be policy-relative remoteness and suppose the unweighted score-times-advantage contribution satisfies $\|g(r)\| \leq C(1+r)^k$ for finite constants $C > 0$ and $k \geq 0$. For any $\lambda > 0$, define $\tilde{g}(r) = e^{-\lambda r}g(r)$. Then $\|\tilde{g}(r)\| \rightarrow 0$ as $r \rightarrow \infty$.*

Proof. The assumption gives

$$\|g(r)\| \leq C(1+r)^k e^{-\lambda r}.$$

For integer $m > k$, the exponential series implies $e^{\lambda r} \geq (\lambda r)^m / m!$ for $r > 0$. Hence

$$(1+r)^k e^{-\lambda r} \leq \frac{m!(1+r)^k}{(\lambda r)^m},$$

and the right-hand side converges to zero because $m > k$. The squeeze theorem proves the result. The proposition controls the weighted gradient tail; it makes no assumption that sample utility decays exponentially. $\square$

Evidence and claim boundary. The empirical intervention establishes a controlled C-U1 causal transmission result. The proposition establishes only a tail bound under finite-order score growth. Neither component asserts that exponential weighting is universally superior, and the proposition is not a utility-decay model.

Pipeline audit

Every number in the result paragraphs is rendered from `research_snapshot.json`. The snapshot verifies the registered status, artifact checksums, held-out seed count, terminal audit, and terminology before any prose or figure is produced. A missing input fails closed rather than creating an empty table or placeholder result.